Q.31 Determine the centroid of a triangle and explain its location with diagram.

Q.32 Define ideal machine, reversible machine, and self-locking machines with examples.

Q.33 Explain the first system of pulleys and derive its velocity ratio.

Q.34 State and explain the torsion equation and its terms.

Q.35 Derive the formula for power transmitted by a shaft.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Classify force systems and explain in detail:

- Coplanar force
- Non- coplanar forces
- Concurrent and parallel forces

Q.37 Two equal and opposite forces of 100N act at a distance of 0.6m apart. Find the moment of the couple and explain its effects.

Q.38 A rectangular lamina is 20cm long and 10cm wide. If a 5cm x 5 cm square is removed from top left corner, find the centroid of the remaining lamina

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Sub : Applied Mechanics

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Which is a vector quantity?

a) Distance b) Speed

c) Force d) Time

Q.2 Law of superposition applies to:

a) Single force b) Multiple forces

c) Energy d) Work

Q.3 Parallel forces may be:

a) Like b) Unlike

c) Both d) None

Q.4 Moment depends on:

a) Force only b) Distance only

\ c) Both d) Time

Q.5 CG, depends on:

a) Shape and mass distribution

b) Color

c) Temperature

d) Time

Q.6 Static friction acts when the body is:
a) Moving b) At rest
c) Falling d) Rotating

Q.7 Friction is a:
a) Conservative force
b) Elastic force
c) Zero force
d) Non conservative force

Q.8 A reversible machine has efficiency:
a) Greater than 50% b) Less than 50%
c) Equal to zero d) Infinite

Q.9 Torsion causes:
a) Linear deformation b) Angular deformation
c) No deformation d) Compression

Q.10 Shear stress is proportional to:
a) Radius b) Length
c) Area d) Volume

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 SI is an international system of unit. (True / False)

Q.12 State Lami's theorem.

Q.13 What are concurrent forces?

Q.14 Define load arm.

Q.15 The maximum value of static friction is called ____ friction.

Q.16 The coefficient of friction has ____ unit.

Q.17 What is the distance of centroid of a triangle from its base?

Q.18 Mechanical Advantages (MA) is the ratio of ____ to efforts.

Q.19 A machines is self-locking if its efficiency is less than_____.

Q.20 Unit of modulus of rigidity is Newton. (True / False)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Differentiate between scalar and vector quantities with examples.

Q.22 State and explain the polygon law of forces.

Q.23 Define equilibrant force and explain how it is determined.

Q.24 A force of 100N acts at an angle of 30º to the horizontal. Resolve it into horizontal and vertical components.

Q.25 Explain the working principle of a simple lever with diagram.

Q.26 Distinguish between a force and a couple.

Q.27 A lever is 2m long. A load of 100N is placed at one end. Find the effort required at the other end to balance the lever.

Q.28 Compare static friction and kinetic friction with examples.

Q.29 Discuss the advantages and disadvantages of friction in daily life.

Q.30 Explain the properties of centre of gravity.